Exp. 11 Testing in Azure DevOps

AIM

To create and execute test cases using Azure DevOps Test Plans and integrate automated testing into a CI/CD pipeline for validating application quality.

TOOLS REQUIRED

• Microsoft Azure DevOps
• GitHub or Azure Repos
• Web Browser
• Azure Subscription

PREREQUISITES

• Project source code available in GitHub
• Azure DevOps account and project created
• CI/CD pipeline already configured
• Application deployed or build ready

PROCEDURE

Step 1: Create Test Plan

• Login to Azure DevOps portal
• Select your project
• Navigate to Test Plans
• Click New Test Plan
• Enter:
o Name: *QuizApp-TestPlan*
o Area Path: Select project
o Iteration: Current sprint
• Click Create

Step 2: Create Test Suite

• Inside Test Plan → Click New Test Suite
• Select:
o Static Suite
• Enter Suite Name:
o *Login Module Testing*

Step 3: Add Test Cases

• Click New Test Case
• Enter details:

Example Test Case:

Field	Value
Title	Verify Login Functionality
Steps	1. Enter username 2. Enter password 3. Click Login
Expected Result	User should login successfully

• Save and Close

Step 4: Execute Test Cases

• Go to Test Plans → Run
• Select test suite
• Click Run for web application
• Perform steps manually
• Mark result:
o Pass / Fail

Step 5: View Test Results

• Navigate to Test Plans → Results
• Analyze:
o Passed tests
o Failed tests
o Bugs (if any)

Step 6: Integrate Automated Testing in Pipeline

Update YAML pipeline to include testing:

```
trigger:
- main

pool:
  vmImage: 'ubuntu-latest'

steps:
- task: UseNode@1
```

```
  inputs:
    version: '18.x'

- script: |
    npm install
    npm run test
  displayName: 'Run Tests'

- script: |
    npm run build
  displayName: 'Build Application'

- task: AzureWebApp@1
  inputs:
    azureSubscription: 'Your-Azure-Connection'
    appName: 'quizapp-online'
    package: '.'
```

Step 7: Run Pipeline with Tests

• Go to Pipelines → Run Pipeline
• Pipeline will:
o Install dependencies
o Execute test cases
o Build application
o Deploy if tests pass

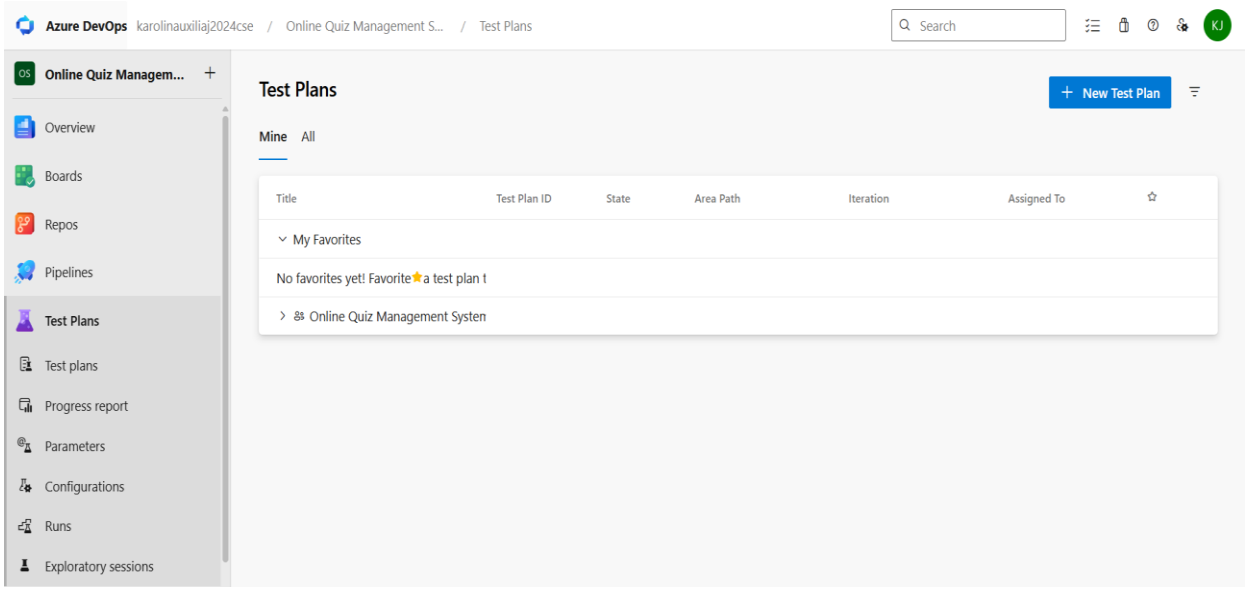
Step 8: Monitor Test Execution

• Go to Pipelines → Runs
• Check:
o Test logs
o Pass/Fail status
• Fix errors if tests fail

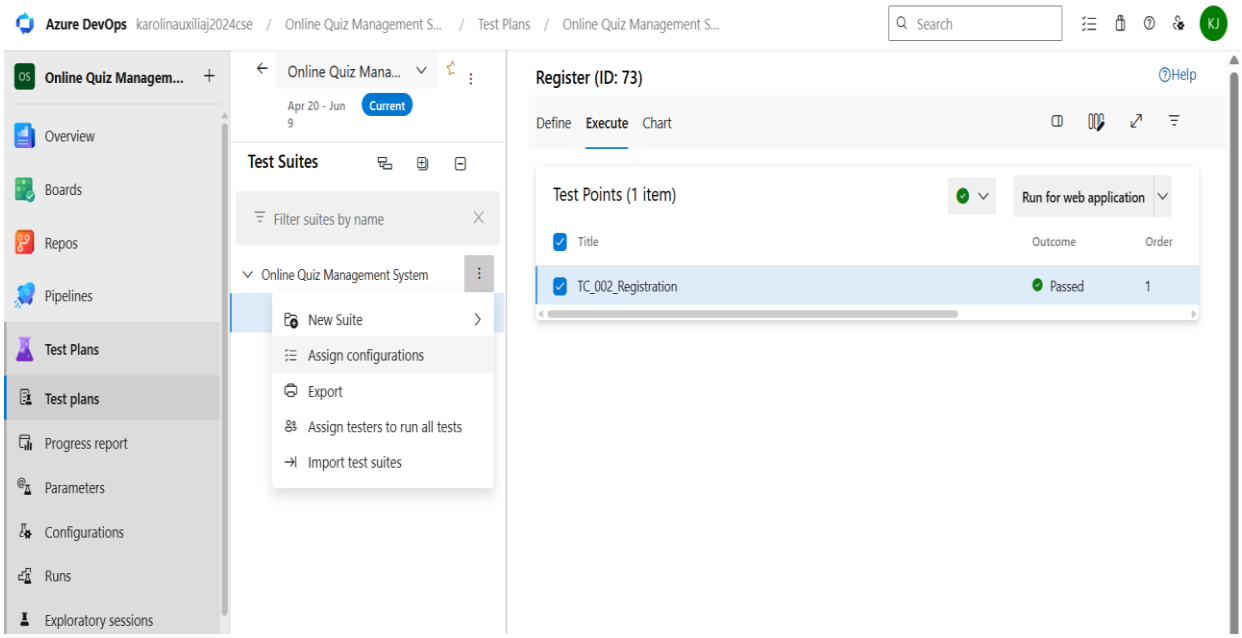
RESULT

Testing was successfully implemented using Azure DevOps Test Plans and automated testing in the CI/CD pipeline. The application quality was validated before deployment, ensuring reliable releases.

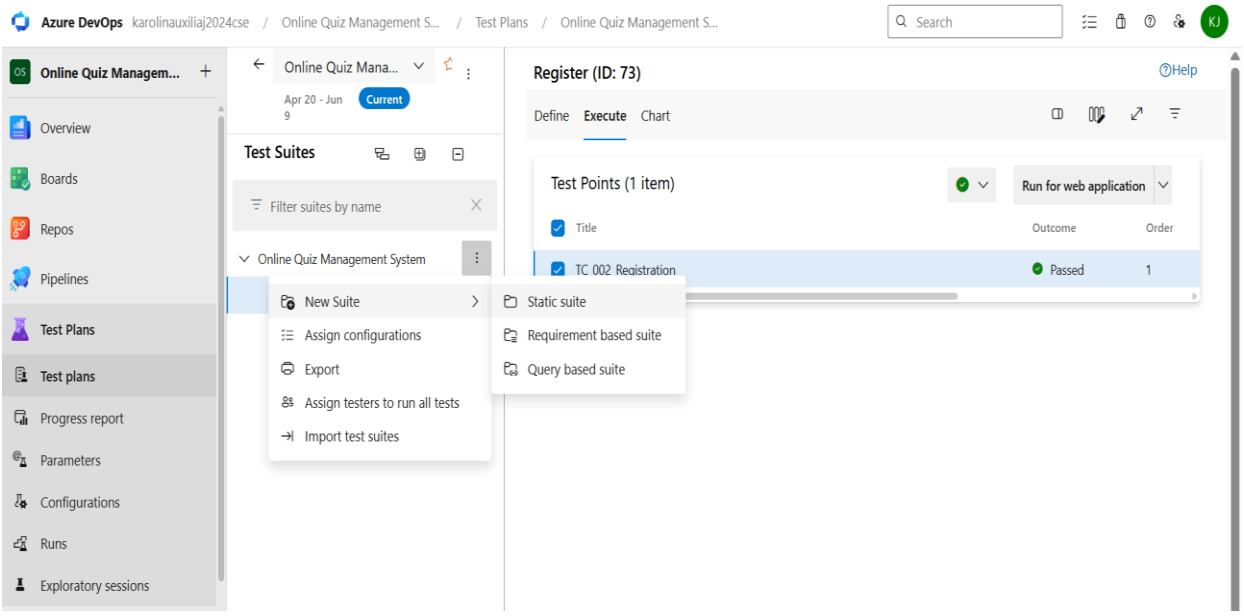
Test Plan



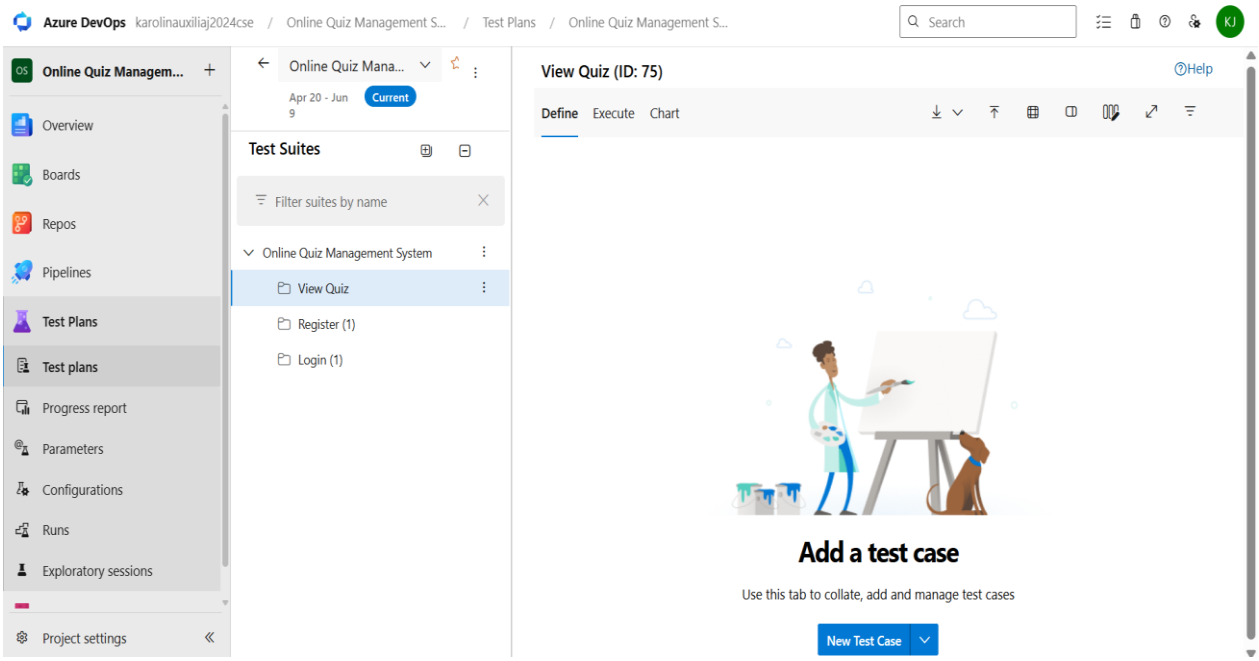
Click New Suite



Click Static Suite



Name as View quiz



NEW TEST CASE *

TC_003_ViewAvailableQuizzes

Karolin Auxilia J | 0 Comments | Add Tag | Save and Close

State: Design | Area: Online Quiz Management System

Reason: New | Iteration: Online Quiz Management System\Sprint 7

Steps | Summary | Associated Automation | 0 | 0

Steps

Steps	Action	Expected result
1.	Login and go to dashboard; View quizzes list; Select a quiz	Quiz list displayed with required details; User select quiz

Click or type here to add a step

Recent test results

Run manual tests with web runner. Learn how to get started

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

NEW TEST CASE *

TC_003_ViewAvailableQuizzes

Karolin Auxilia J | 0 Comments

State: Design | Area:

Reason: New | Iteration:

Discussion

Add a comment. Use # to link a work item, @ to men

Add link

You are adding a link from:

0 TC_003_ViewAvailableQuizzes
Design

Link type:

Parent

Work items to link 1 | Clear all

Search work items by ID or title

- Test Suite 70: Online Quiz Management System
- Test Suite 73: Register
- Test Case 74: TC_002_Registration
- Test Suite 71: Login
- Test Case 72: TC_001_Login
- Test Plan 69: Online Quiz Management System
- User Story 62: seat allocation
- User Story 63: Bus pass application

Search

Save and Close

Summary | Associated Automation | 0 | 0

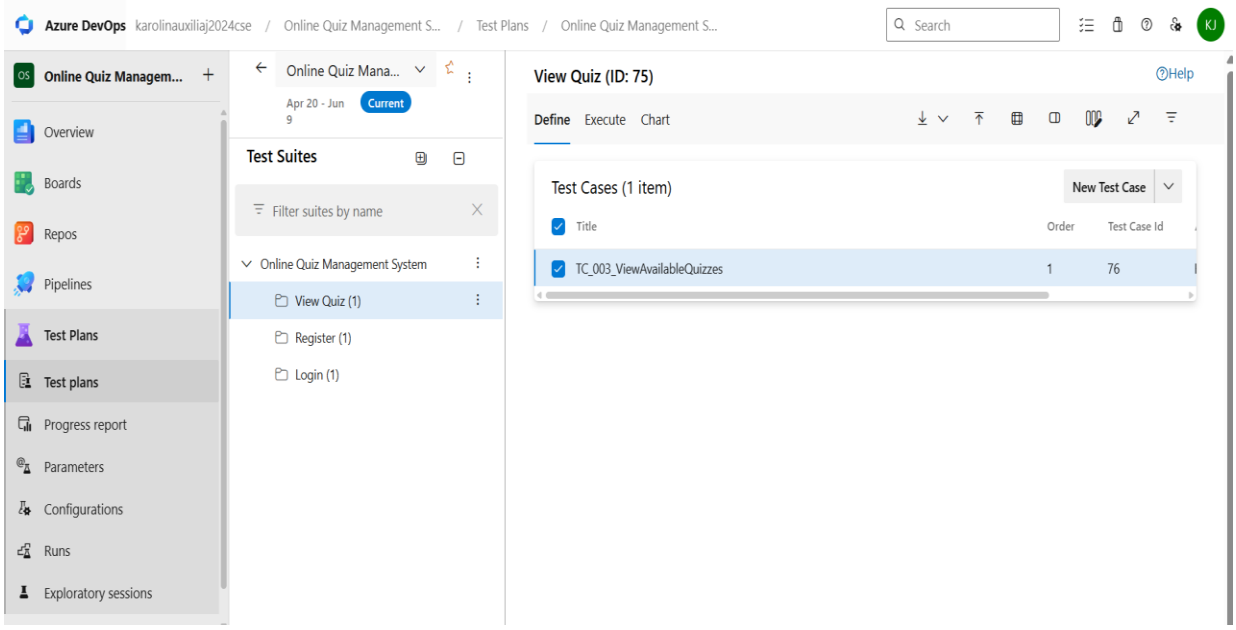
d link

ink an Azure Repos commit, pull request or branch to ee the status of your development. You can also create branch to get started.

ed Work

d link

d an existing work item as a parent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Search
Online Quiz Managem...
Overview
Boards
Repos
Pipelines
Test Plans
Test plans
Progress report
Parameters
Configurations
Runs
Exploratory sessions
Online Quiz Mana...
Apr 20 - Jun 9
Current
Test Suites
Filter suites by name
Online Quiz Management System
View Quiz (1)
Register (1)
Login (1)
View Quiz (ID: 75)
Help
Define
Execute
Chart
Test Cases (1 item)
New Test Case
Title
Order
Test Case Id
TC_003_ViewAvailableQuizzes
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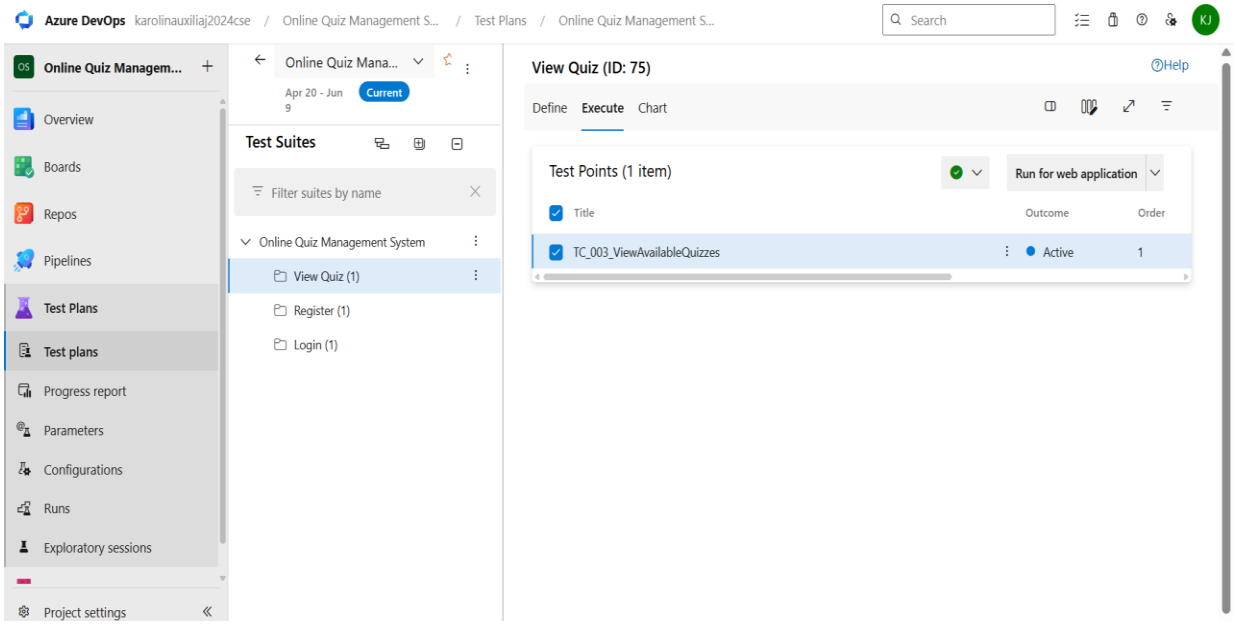
76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Test Points (1 item)
Run for web application
Title
Outcome
Order
TC_003_ViewAvailableQuizzes
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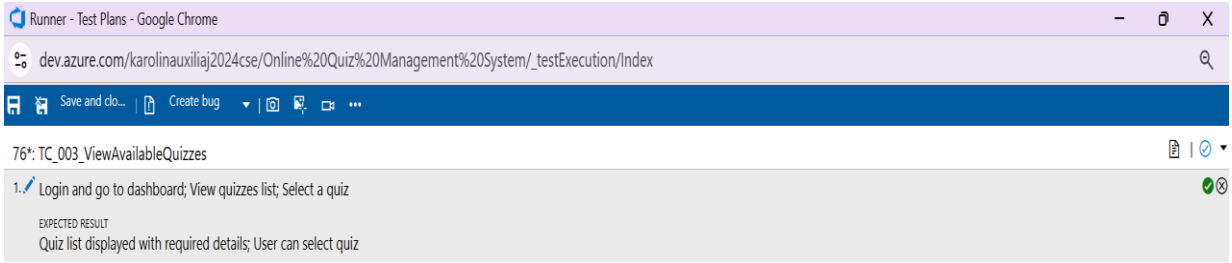
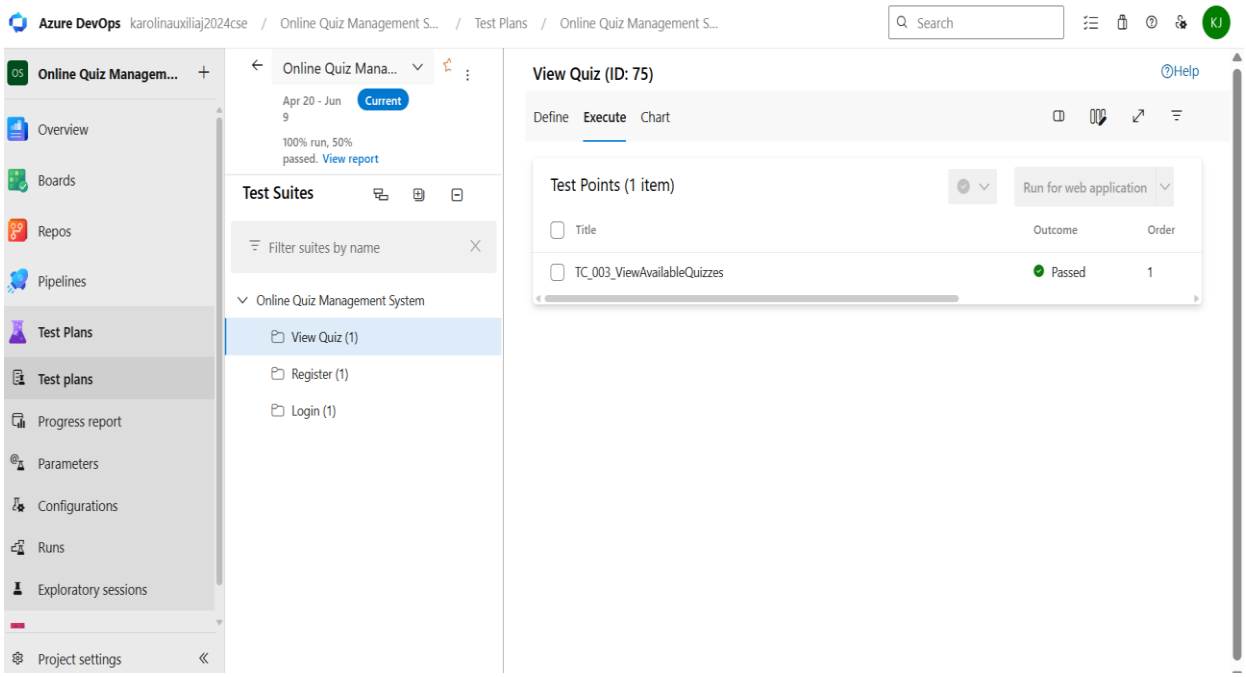

Runner - Test Plans - Google Chrome
dev.azure.com/karolinauxiliaj2024cse/Online%20Quiz%20Management%20System/_testExecution/Index
Save and clo...
Create bug
76*: TC_003_ViewAvailableQuizzes
1. Login and go to dashboard; View quizzes list; Select a quiz
EXPECTED RESULT
Quiz list displayed with required details; User can select quiz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Search
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Current
100% run, 50% passed. View report
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Passed
1